

- 6 Cloves are the aromatic flower buds of the tree *Syzygium aromaticum*. Eugenol is a drug extracted from cloves.

Eugenol affects the movement of sodium ions through the cell surface membranes of sensory neurones.

- (a) Fig. 6.1 shows the effect of eugenol concentration on the percentage decrease in sodium ion movement.

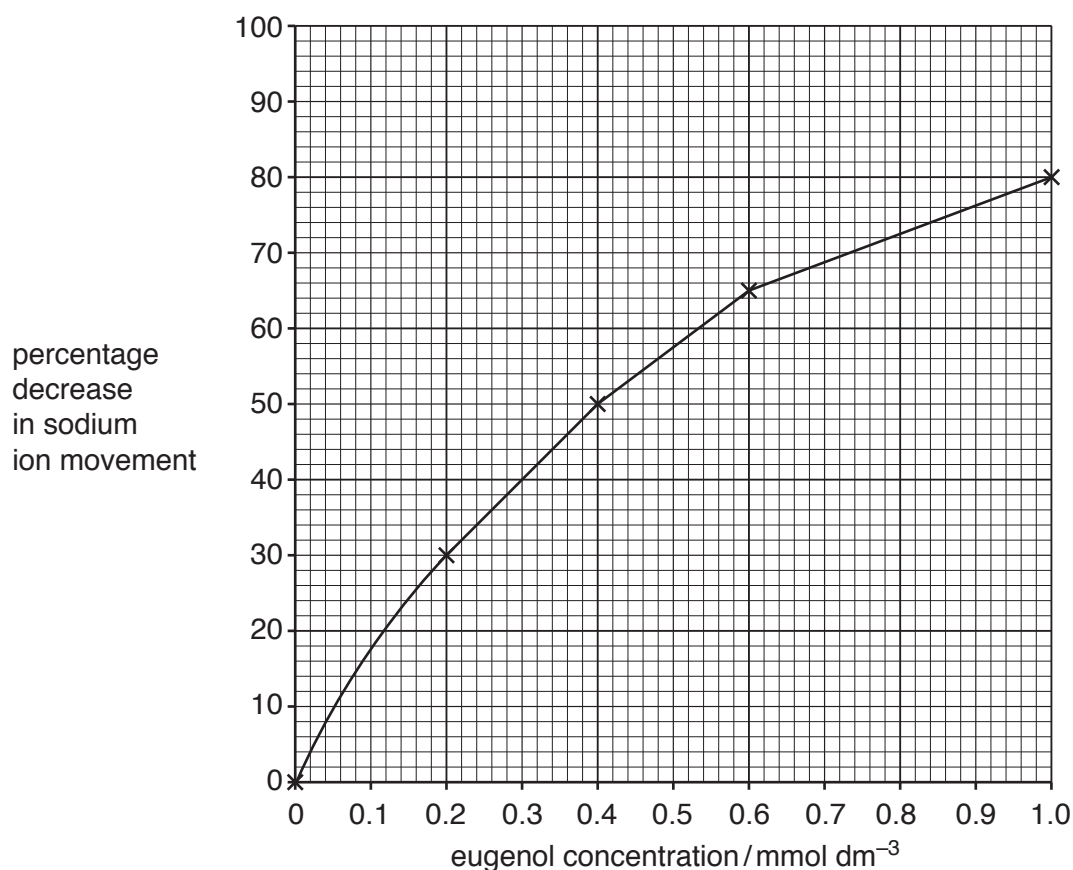


Fig. 6.1

- (i) Describe the effect of eugenol concentration on the percentage decrease in sodium ion movement.

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- (ii) Using Fig. 6.1, determine the percentage decrease in sodium ion movement at a eugenol concentration of 0.5 mmol dm^{-3} .

percentage decrease % [1]

- (b)** Eugenol can be used to relieve the pain of toothache by putting it onto the affected tooth and gum.

Suggest **and** explain how the action of eugenol may reduce pain.

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[Total: 8]